

Navigators Technical Documentation

Navigators: Continuous Smartphone-Based Vehicle Dead Reckoning via Causal Temporal Convolutions, Kinematic Constraints, and 15-State Extended Kalman Filtering

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Abstract

Global Navigation Satellite System (GNSS) degradation or total outage poses severe reliability hazards for consumer vehicular navigation in urban canyons, road tunnels, and multi-level infrastructure. While tactical-grade Inertial Navigation Systems (INS) achieve centimeter-scale accuracy using high-precision ring-laser gyroscopes and vehicle wheel odometry, smartphone-based navigation is constrained to low-cost Micro-Electro-Mechanical Systems (MEMS) sensors mounted arbitrarily inside the vehicle cabin without access to CAN bus telemetry or external wheel encoders.

This paper presents **Navigators**, a fully autonomous, edge-executable intelligent dead-reckoning architecture designed to maintain continuous 2D horizontal trajectory estimation during prolonged GNSS outages using commercial off-the-shelf smartphones. The proposed method integrates: (1) an automated Triad-based leveling and heading alignment procedure mapping arbitrary device attitudes into the vehicle dynamic frame; (2) a strictly causal Temporal Convolutional Network (TCN) with 5.40 million parameters that predicts planar vehicle velocity from windowed 6-axis IMU signals without future-temporal leakage; (3) a 15-state Extended Kalman Filter (EKF) formulated in the local East-North-Up (ENU) frame fusing AI velocity pseudo-measurements, Non-Holonomic Constraints (NHC), and Zero-Velocity Updates (ZUPT); and (4) an offline vector road network geometric map matcher. The complete edge navigation pipeline executes entirely in client-side WebAssembly via SIMD vectorization with sub-2-millisecond processing latency per 10 Hz cycle. Empirical software validation across 288 automated test suites demonstrates mathematical stability, zero data leakage, and verified PyTorch-to-ONNX numerical parity. Physical in-vehicle road testing under natural GNSS attenuation is characterized as ongoing experimental work.

Keywords: Dead Reckoning, Inertial Navigation, Extended Kalman Filter, Temporal Convolutional Network, Non-Holonomic Constraints, Sensor Fusion, Edge WebAssembly, GNSS Outage.

1. Introduction

Satellite-based navigation has become the foundational utility for global surface transportation. However, satellite microwave signals (L-band) are fundamentally vulnerable to line-of-sight obstruction, atmospheric refraction, multi-path reflections from glass and steel facades, and complete signal attenuation inside subterranean tunnels and multi-level parking structures.

In automotive engineering, conventional dead reckoning relies on fused sensor suites comprising Anti-lock Braking System (ABS) wheel speed sensors, transmission gear encoders, steering angle sensors, and chassis-welded tactical IMUs connected via the Controller Area Network (CAN) bus. Conversely, millions of commercial rideshare drivers, delivery fleets, and everyday motorists navigate exclusively using consumer smartphones placed on windshield suction mounts or magnetic dashboard clips. These devices lack any physical or digital interface to vehicle odometry, forcing position estimators to rely solely on uncalibrated, noisy, consumer-grade MEMS accelerometers and gyroscopes.

Navigators addresses this fundamental technical challenge by establishing an end-to-end mathematical and neural dead-reckoning system capable of running locally on a smartphone without requiring cloud server connectivity, vehicle CAN bus integration, or active satellite signals.

2. Problem Statement

Let $\mathbf{p}(t) \in \mathbb{R}^3$ denote the true position of a vehicle in the Earth-Centered Earth-Fixed (ECEF) or local tangent frame. At time t_{outage} , satellite reception is severed:

$$\mathbf{z}_{gnss}(t) = \emptyset, \quad \forall t \geq t_{outage}$$

The smartphone provides only noisy specific force measurements $\mathbf{f}^b(t)$ and angular rates $\boldsymbol{\omega}^b(t)$ from its internal MEMS triad at an unaligned device coordinate frame $\{b\}$:

$$\mathbf{f}^b(t) = \mathbf{R}_v^b(t) \mathbf{a}^v(t) + \mathbf{R}_n^b(t) \mathbf{g}^n + \mathbf{b}_a(t) + \mathbf{w}_a(t)$$

$$\boldsymbol{\omega}^b(t) = \mathbf{R}_v^b(t) \boldsymbol{\omega}^v(t) + \mathbf{b}_g(t) + \mathbf{w}_g(t)$$

Under naive direct integration, uncorrected accelerometer bias $\mathbf{b}_a$ and gyroscope bias $\mathbf{b}_g$ induce quadratic and cubic position errors:

$$\Delta \mathbf{p}(t) \approx \frac{1}{2} \mathbf{b}_a t^2 + \frac{1}{6} \mathbf{g} \times \mathbf{b}_g t^3$$

On smartphone MEMS hardware where bias instability ranges from 0.1 to 0.5 m/s², pure integration yields over 500 meters of position error within 60 seconds, rendering navigation unusable.

The objective of Navigators is to estimate the horizontal trajectory $\hat{\mathbf{p}}_{\text{ENU}}(t) = [\hat{p}_E(t), \hat{p}_N(t)]^T$ such that position drift is bounded throughout the outage interval $[t_{outage}, t_{outage} + T]$ without external odometry.

3. Research Motivation

- 1. Ubiquity of Smartphone Navigation:** Requiring proprietary OBD-II dongles or vehicle CAN bus integration limits dead-reckoning solutions to modern or commercial fleet vehicles. A software-only smartphone architecture democratizes safety and navigation continuity.

- 2. **True Operational Autonomy:** Cloud-based deep learning inference fails in tunnels due to concurrent cellular blackouts. True resilience requires the entire AI model and sensor fusion pipeline to execute locally on the device edge.
- 3. **Strict Causal Realism:** Prior academic literature in AI dead reckoning frequently suffers from subtle data leakage, including bidirectional LSTM layers looking into future timesteps, acausal Gaussian window filtering across outage boundaries, or random splitting of overlapping time windows. Navigators establishes an empirically rigorous framework with strict causal separation.

4. Related Technical Background

Traditional inertial navigation systems (INS) operate by integrating specific force and angular rate equations through mechanization algorithms (Titterton & Weston, 2004; Groves, 2013). In land vehicles, kinematic non-holonomic constraints (NHC) first formulated by Dissanayake et al. (2001) exploit the physical reality that wheeled vehicles do not slide sideways or bounce vertically under nominal conditions. Zero-Velocity Updates (ZUPT) detected via generalized likelihood ratio tests or acceleration variance thresholds (Skog et al., 2010) provide essential periodic recalibration during stationary stops.

In recent years, data-driven velocity estimation using deep neural networks (Abe et al., 2019; Corti et al., 2020) has emerged to replace physical wheel odometry by learning the subtle vibration signatures of vehicular tires, engine harmonics, and suspension dynamics directly from high-rate IMU time-series.

5. Existing Approaches & Research Gap

Approach	External Sensors Required	Offline Edge Capable	CAN Bus Free	Causal Guarantees	Drift Mitigation
Traditional Tactical INS	High-grade FOG/RLG	Yes	Yes	Yes	High hardware cost (> \$10,000)
Automotive OEM DR	Wheel Encoders / OBD-II	Yes	No	Yes	Restricted to vehicle hardware
Cloud-Based AI DR	Smartphone MEMS	No	Yes	Often Acausal	Fails in subterranean tunnels
Navigators (This Work)	Consumer Smartphone MEMS	Yes (Wasm)	Yes	Strictly Causal	TCN + 15-State EKF + NHC + Map Snapping

The Research Gap:

Existing smartphone dead-reckoning research rarely couples causal, edge-executable neural velocity regression with a full 15-state Extended Kalman Filter, local vector map matching, and a deterministic state machine within a client-side WebAssembly browser architecture. Furthermore, experimental reproducibility and zero-leakage session partitioning are frequently overlooked in published benchmarks.

6. Proposed Method

Navigators implements a modular, six-stage estimation pipeline:



7. Mathematical System Formulation

7.1 Sensor Kinematics & Coordinate Alignment

Let $\{b\}$ denote the arbitrary phone body frame, $\{v\}$ the vehicle frame, and $\{n\}$ the local ENU navigation frame. The rotation from phone to vehicle $\mathbf{R}_b^v$ is resolved via the Triad algorithm during pre-outage dwell and initial forward motion:

$$\mathbf{u}_{\text{down}}^b = \frac{\|\mathbf{f}\|_{\text{dwell}}^b}{\|\mathbf{f}\|_{\text{dwell}}^b} \mathbf{f}_{\text{dwell}}^b$$

$$\mathbf{u}_{\text{fwd}}^b = \frac{\mathbf{a}_{\text{motion}}^b - (\mathbf{a}_{\text{motion}}^b \cdot \mathbf{u}_{\text{down}}^b) \mathbf{u}_{\text{down}}^b}{\|\mathbf{a}_{\text{motion}}^b - (\mathbf{a}_{\text{motion}}^b \cdot \mathbf{u}_{\text{down}}^b) \mathbf{u}_{\text{down}}^b\|}$$

$$\mathbf{R}_b^v = \left[\frac{\mathbf{u}_{\text{fwd}}^b \times \mathbf{u}_{\text{down}}^b}{\|\mathbf{u}_{\text{fwd}}^b \times \mathbf{u}_{\text{down}}^b\|}, \frac{\mathbf{u}_{\text{fwd}}^b}{\|\mathbf{u}_{\text{fwd}}^b\|}, \frac{\mathbf{u}_{\text{down}}^b}{\|\mathbf{u}_{\text{down}}^b\|} \right]^T$$

7.2 15-State Vector Definition

$$\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T, \boldsymbol{\theta}^T, \mathbf{b}_a^T, \mathbf{b}_g^T]^T \in \mathbb{R}^{15}$$

Propagated at discrete timestep $\Delta t = 0.1$ s via standard error-state mechanization:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{f}(\hat{\mathbf{x}}_{k-1}, \mathbf{f}_k^b, \boldsymbol{\omega}_k^b)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^T + \mathbf{Q}_k$$

7.3 Multi-Constraint Updates

1. **AI Velocity Pseudo-Measurement:** $\mathbf{z}_{ai} = [v_E^{tcn}, v_N^{tcn}]^T$, $\mathbf{R}_{ai} = \text{diag}(0.3^2, 0.3^2)$.
2. **Non-Holonomic Constraints (NHC):** Enforces $v_{lateral}^V = 0$ and $v_{vertical}^V = 0$ in the vehicle body frame:

$$\begin{aligned} \mathbf{z}_{nhc} &= [0, 0]^T, \quad \mathbf{H}_{nhc} = \begin{bmatrix} -\sin\psi & \cos\psi & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{R}_{nhc} = \text{diag}(1.0, 1.0) \\ \mathbf{z}_{zupt} &= [0, 0, 0]^T, \quad \mathbf{R}_{zupt} = \text{diag}(0.01, 0.01, 0.01) \end{aligned}$$

3. **Zero-Velocity Updates (ZUPT):** Applied when acceleration variance $\sigma_a^2 < 0.05 \text{ m}^2/\text{s}^4$ and angular rate norm $\|\boldsymbol{\omega}\| < 0.05 \text{ rad/s}$ across a 1.0 s window:

$$\mathbf{z}_{zupt} = [0, 0, 0]^T, \quad \mathbf{R}_{zupt} = \text{diag}(0.01, 0.01, 0.01)$$

4. **Joseph-Form Stabilized Covariance:**

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_{k|k-1}(\mathbf{I} - \mathbf{K}\mathbf{H})^T + \mathbf{K}\mathbf{R}\mathbf{K}^T$$

8. Neural Network Velocity Estimator

The neural estimator is a custom Temporal Convolutional Network (`TCNVelocityEstimator`) featuring 4 residual blocks with left-padded causal 1D convolutions: - **Input Channels:** 6 ($a_x, a_y, a_z, \omega_x, \omega_y, \omega_z$). - **Window Size:** 200 timesteps (20.0 s at 10 Hz). - **Dilation Schedule:** $d \in \{1, 2, 4, 8\}$ with kernel size $K = 7$, yielding an effective receptive field of 181 timesteps. - **Normalization:** Z-score standardization applied per window using training set statistics:

$$\hat{\mathbf{X}} = \frac{\mathbf{X} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}$$

- **Parameter Count:** 5,400,322 weights exported to ONNX (float32, 6.78 MB).

9. Experimental Dataset & Preprocessing

9.1 Dataset Provenance

Training and validation rely on the public benchmark **IO-VNBD** (Inertial and Odometry Vehicle Navigation Benchmark Dataset; Onyekpeu et al.): - Synchronized sensor (`S_*.csv`) and ground-truth (`V_*.csv`) CSV pairs. - Downsampled with anti-aliasing to 10 Hz.

9.2 Session-Level Data Partitioning

To guarantee zero temporal leakage, sliding windows are never shuffled across splits: - **Train Set:** 100 independent sessions (63,423 windows). - **Validation Set:** 22 independent sessions (30,349 windows). - **Test Set:** 22 independent sessions (11,934 windows).

10. Empirical Results & Findings

10.1 Software Architecture & Numerical Parity

- **Automated Test Suite: 288 / 288 test suites passed** (274 pytest backend suites + 14 Node.js engine suites).
- **ONNX Export Parity:** Validated against PyTorch across all test windows within absolute tolerance $\text{atol} = 10^{-5}$ and relative tolerance $\text{rtol} = 10^{-4}$.
- **Edge Inference Latency:** Measured at 0.7 ms on Apple Silicon M-series via WebAssembly SIMD (`ort-wasm-simd-threaded.wasm`).

10.2 Seven-Stage Ablation Findings (Simulated / Replay Suite)

Evaluations using `replay.py` across identical outage windows illustrate the role of each architectural layer: - **Mode A (Raw IMU):** Diverges within 15 seconds; quadratic drift exceeds 500 m. - **Mode B (Filtered IMU):** Mitigates high-frequency vibration spikes but maintains baseline integration drift. - **Mode C (AI Velocity):** Replaces acceleration double-integration with single-integration of predicted velocity; drift rate becomes linear rather than quadratic. - **Mode D (AI + EKF):** Actively tracks and corrects sensor bias; stabilizes attitude estimation. - **Mode E (AI + EKF + NHC):** Suppresses transverse skid and unphysical vertical lift. - **Mode F (AI + EKF + NHC + ZUPT):** Completely halts drift during traffic stops and stationary dwells. - **Mode G (Full System):** Eliminates long-term cross-track angular drift by snapping coordinates to valid road centerlines.

11. Discussion & Limitations

1. **Physical Road Trial Status:** While the software implementation, mathematical formulation, and edge WebAssembly engines are fully realized and verified in automated tests, **physical road trials with a real moving smartphone inside an actual tunnel remain pending.**
2. **Thermal & Operating System Throttling:** On lower-tier mobile hardware, prolonged continuous execution of WebAssembly neural inference and high-rate sensor capture may induce thermal throttling or background execution suspension by aggressive mobile operating systems.
3. **Turn-Rich Unmapped Environments:** In off-road terrain or unmapped parking structures where road centerline vectors are unavailable, the system operates purely on Mode F (AI + EKF + NHC + ZUPT), which accumulates heading drift over extended denial periods.

12. Reproducibility Statement

All source code, configuration files, model definitions, replay evaluators, and test fixtures are fully available within this repository: - Configuration: `configs/default.yaml` - Replay Harness: `replay.py` - Test Runner: `pytest tests` and `node --test tests/*.test.cjs` - Normalization Contract: `simulator/norm_stats.json`

13. Conclusion

Navigators demonstrates a rigorous, software-only dead-reckoning architecture for consumer smartphones navigating through GNSS outages. By combining causal Temporal Convolutional Networks, a 15-state Extended Kalman Filter, kinematic Non-Holonomic Constraints, and offline vector map matching within a

lightweight WebAssembly client, the system establishes a robust blueprint for resilient land navigation without external odometry or cloud dependencies.

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